

Basic Biochemistry - 1 Mark Question Notes

Based on the uploaded syllabus and Biochemistry source PDF | Total Questions: 130

How to use: Read the answer first, then cover it and recall the keyword. These are short, exam-focused 1-mark points from the full Basic Biochemistry syllabus.

Syllabus Coverage

Unit	Topics Covered
Unit I - Carbohydrate Chemistry	Definition, classification, examples, glycolysis, gluconeogenesis, TCA cycle, energetics, blood glucose regulation
Unit II - Lipid Chemistry	Definition, functions, classification, fatty acids, essential fatty acids, phospholipids
Unit III - Amino Acids, Peptides and Proteins	Amino acids, D/L structure, peptides, important peptides, protein classification and functions
Unit IV - Nucleotides and Nucleic Acids	Nucleotides, DNA, RNA, Watson-Crick model, tRNA, rRNA, mRNA
Unit V - Enzymes	Definition, classification, active site, cofactors, coenzymes, proenzymes, mechanism
Unit VI - Vitamins and Minerals	Vitamin classification, functions, disorders; minerals and their functions
Unit VII - Nutrition, Acid-Base and Buffers	Balanced diet, caloric values, RDA, BMR, SDA, acids, bases, pH, buffer, acid-base balance

Unit I - Carbohydrate Chemistry

No.	1-Mark Question	Answer / Key Point
1	Define carbohydrates.	Carbohydrates are polyhydroxy aldehydes or ketones, or compounds that yield them on hydrolysis.
2	What are the three main classes of carbohydrates?	Monosaccharides, oligosaccharides and polysaccharides.
3	Give two examples of monosaccharides.	Glucose and fructose.
4	Give two examples of disaccharides.	Sucrose and lactose.
5	Give two examples of polysaccharides.	Starch and glycogen.
6	What is glycolysis?	Glycolysis is the pathway that converts glucose into pyruvate or lactate with production of ATP.
7	Where does glycolysis occur in the cell?	In the cytosol.
8	Name the rate-limiting enzyme of glycolysis.	Phosphofructokinase-1 (PFK-1).
9	What is the end product of aerobic glycolysis?	Pyruvate.
10	What is the end product of anaerobic glycolysis?	Lactate.

No.	1-Mark Question	Answer / Key Point
11	What is the net ATP gain in anaerobic glycolysis?	2 ATP per glucose.
12	What is gluconeogenesis?	Gluconeogenesis is the synthesis of glucose from non-carbohydrate precursors.
13	Name two important precursors of gluconeogenesis.	Lactate, glycerol and glucogenic amino acids.
14	Name one tissue where gluconeogenesis mainly occurs.	Liver; kidney also contributes during prolonged fasting.
15	Name the enzyme that converts glucose-6-phosphate into glucose.	Glucose-6-phosphatase.
16	What is the citric acid cycle also called?	Krebs cycle or TCA cycle.
17	Where does the citric acid cycle occur?	In the mitochondrial matrix.
18	What condenses with oxaloacetate to form citrate?	Acetyl-CoA.
19	Name the enzyme forming citrate in TCA cycle.	Citrate synthase.
20	How many ATP are produced from one acetyl-CoA in the traditional textbook calculation?	12 ATP.
21	How many ATP are produced by complete oxidation of one glucose in the traditional calculation?	38 ATP.
22	What hormone lowers blood glucose level?	Insulin.
23	Name two hormones that increase blood glucose level.	Glucagon and adrenaline.
24	What is the normal fasting blood glucose range?	About 70-110 mg/dL.

Unit II - Lipid Chemistry

No.	1-Mark Question	Answer / Key Point
25	Define lipids.	Lipids are a heterogeneous group of water-insoluble organic compounds, soluble in non-polar solvents.
26	Mention two major functions of lipids in the body.	They store energy and form structural components of cell membranes.
27	How are lipids classified based on composition?	Simple lipids, compound lipids and derived lipids.
28	Give one example of a simple lipid.	Triacylglycerol or wax.
29	Give one example of a compound lipid.	Phospholipid or glycolipid.
30	Give one example of a derived lipid.	Cholesterol or fatty acid.
31	Define fatty acids.	Fatty acids are carboxylic acids with hydrocarbon chains.
32	Classify fatty acids based on saturation.	Saturated and unsaturated fatty acids.

No.	1-Mark Question	Answer / Key Point
33	Give one example of a saturated fatty acid.	Palmitic acid or stearic acid.
34	Give one example of an unsaturated fatty acid.	Oleic acid, linoleic acid or arachidonic acid.
35	What are essential fatty acids?	Fatty acids that cannot be synthesized in the body and must be supplied in diet.
36	Name two essential fatty acids.	Linoleic acid and alpha-linolenic acid.
37	What are phospholipids?	Compound lipids containing phosphoric acid, alcohol, fatty acid and usually nitrogenous base.
38	Why are phospholipids important?	They are major components of biological membranes and help in lipoprotein formation.
39	Give two examples of phospholipids.	Lecithin and cephalin.

Unit III - Amino Acids, Peptides and Proteins

No.	1-Mark Question	Answer / Key Point
40	Define amino acids.	Amino acids are organic compounds containing amino and carboxyl groups.
41	What is the general structure of an alpha-amino acid?	A central alpha-carbon attached to -NH ₂ , -COOH, H and an R group.
42	How are amino acids classified nutritionally?	Essential and non-essential amino acids.
43	Give two examples of essential amino acids.	Lysine and tryptophan.
44	Give two examples of non-essential amino acids.	Alanine and glycine.
45	What are acidic amino acids?	Amino acids with an extra carboxyl group; examples are aspartate and glutamate.
46	What are basic amino acids?	Amino acids with extra basic groups; examples are lysine, arginine and histidine.
47	What is the difference between D and L amino acids?	They differ in spatial arrangement around the asymmetric alpha-carbon.
48	Which amino acid has no asymmetric carbon?	Glycine.
49	Define peptide bond.	A peptide bond is the amide linkage between the carboxyl group of one amino acid and amino group of another.
50	Define peptide.	A peptide is a short chain of amino acids linked by peptide bonds.
51	Give two biologically important peptides.	Glutathione and oxytocin.
52	How are proteins classified based on composition?	Simple, conjugated and derived proteins.
53	Give one example of a simple protein.	Albumin.

No.	1-Mark Question	Answer / Key Point
54	Give one example of a conjugated protein.	Hemoglobin or glycoprotein.
55	How are proteins classified based on shape?	Fibrous proteins and globular proteins.
56	Give one example of a fibrous protein.	Collagen or keratin.
57	Give one example of a globular protein.	Albumin or hemoglobin.
58	Name the four levels of protein structure.	Primary, secondary, tertiary and quaternary structures.
59	What is the primary structure of protein?	The linear sequence of amino acids in a polypeptide chain.
60	Name two major secondary structures of proteins.	Alpha-helix and beta-pleated sheet.
61	Mention two functions of proteins.	They act as enzymes, structural proteins, transporters, hormones and antibodies.

Unit IV - Nucleotides and Nucleic Acids

No.	1-Mark Question	Answer / Key Point
62	Define nucleotides.	Nucleotides are compounds made of nitrogenous base, pentose sugar and phosphate.
63	Define nucleoside.	A nucleoside contains nitrogenous base and pentose sugar, but no phosphate.
64	Name the two types of nucleic acids.	DNA and RNA.
65	What are purine bases?	Adenine and guanine.
66	What are pyrimidine bases in DNA?	Cytosine and thymine.
67	Which pyrimidine replaces thymine in RNA?	Uracil.
68	What is the sugar present in DNA?	2-deoxyribose.
69	What is the sugar present in RNA?	Ribose.
70	Who proposed the double helix model of DNA?	Watson and Crick.
71	Mention one important feature of DNA structure.	Two antiparallel strands are held by complementary base pairing.
72	What is the function of DNA?	It stores and transmits genetic information.
73	Name three major types of RNA.	mRNA, tRNA and rRNA.
74	What is the function of mRNA?	It carries genetic information from DNA to ribosomes for protein synthesis.
75	What is the function of tRNA?	It carries specific amino acids to the ribosome during translation.
76	What is the function of rRNA?	It forms the structural and catalytic core of ribosomes.

Unit V - Enzymes

No.	1-Mark Question	Answer / Key Point
77	Define enzyme.	An enzyme is a biological catalyst that increases the rate of a biochemical reaction without being consumed.
78	Name the six major classes of enzymes.	Oxidoreductases, transferases, hydrolases, lyases, isomerases and ligases.
79	Give one example of an oxidoreductase.	Lactate dehydrogenase.
80	Give one example of a hydrolase.	Amylase or lipase.
81	Give one example of a ligase.	DNA ligase.
82	What is the active site of an enzyme?	The specific region of enzyme where substrate binds and catalysis occurs.
83	What is a cofactor?	A non-protein component required for enzyme activity.
84	Give two examples of metal ion cofactors.	Zn ²⁺ , Mg ²⁺ or Fe ²⁺ .
85	What is a coenzyme?	An organic, non-protein cofactor usually derived from vitamins.
86	Give one example of a coenzyme.	NAD ⁺ , FAD or coenzyme A.
87	What is a proenzyme?	An inactive precursor of an enzyme that becomes active after specific cleavage.
88	Give one example of a proenzyme.	Pepsinogen or trypsinogen.
89	What is the enzyme-substrate complex?	A temporary complex formed when substrate binds to enzyme.
90	How do enzymes increase reaction rate?	They lower the activation energy of the reaction.
91	Name two models of enzyme action.	Lock-and-key model and induced-fit model.

Unit VI - Vitamins and Minerals

No.	1-Mark Question	Answer / Key Point
92	Define vitamins.	Vitamins are organic compounds required in small amounts for normal metabolism, growth and health.
93	Classify vitamins based on solubility.	Fat-soluble vitamins and water-soluble vitamins.
94	Name the fat-soluble vitamins.	Vitamins A, D, E and K.
95	Name the water-soluble vitamins.	Vitamin C and B-complex vitamins.
96	What is the main biochemical function of vitamin A?	Vision, epithelial integrity and growth.
97	Deficiency of vitamin A causes which disorder?	Night blindness and xerophthalmia.
98	What is the main function of vitamin D?	Regulation of calcium and phosphate metabolism.
99	Deficiency of vitamin D causes which disorder?	Rickets in children and osteomalacia in adults.
100	What is the main function of vitamin E?	Antioxidant protection of cell membranes.
101	Deficiency of vitamin K causes what problem?	Bleeding tendency due to impaired clotting factor synthesis.

No.	1-Mark Question	Answer / Key Point
102	Deficiency of vitamin C causes which disease?	Scurvy.
103	Deficiency of thiamine causes which disease?	Beri-beri.
104	Deficiency of niacin causes which disease?	Pellagra.
105	Deficiency of vitamin B12 causes which anemia?	Megaloblastic anemia/pernicious anemia.
106	What is the chief function of calcium?	Formation of bones and teeth, muscle contraction, nerve transmission and blood clotting.
107	What is the chief function of phosphorus?	Bone formation, ATP, nucleic acids and phospholipids.
108	What is the chief function of sodium?	Maintenance of extracellular fluid volume and nerve impulse transmission.
109	What is the chief function of potassium?	Maintenance of intracellular fluid balance and cardiac function.
110	What is the chief function of iron?	Formation of hemoglobin, myoglobin and cytochromes.
111	What is the chief function of magnesium?	It acts as a cofactor for many ATP-dependent enzymes.
112	What is the chief function of copper?	Iron metabolism and activity of oxidase enzymes.
113	What is the chief function of selenium?	Antioxidant function as a component of glutathione peroxidase.
114	What is the chief function of molybdenum?	Cofactor for enzymes such as xanthine oxidase.

Unit VII - Nutrition, Acid-Base and Buffers

No.	1-Mark Question	Answer / Key Point
115	Define balanced diet.	A diet containing all nutrients in proper proportion to meet body requirements.
116	What is the caloric value of carbohydrate?	4 kcal/g.
117	What is the caloric value of protein?	4 kcal/g.
118	What is the caloric value of fat?	9 kcal/g.
119	What does RDA stand for?	Recommended Dietary Allowance.
120	Define RDA.	RDA is the daily amount of a nutrient recommended to maintain good health.
121	Define BMR.	Basal metabolic rate is the energy required at complete physical and mental rest to maintain vital functions.
122	Mention two factors affecting BMR.	Age, sex, body size, hormones, fever and climate.
123	What is SDA or thermic effect of food?	Increase in metabolic rate after food intake due to digestion, absorption and metabolism.
124	Define acid.	An acid is a substance that donates hydrogen ions.

No.	1-Mark Question	Answer / Key Point
125	Define base.	A base is a substance that accepts hydrogen ions.
126	Define pH.	pH is the negative logarithm of hydrogen ion concentration.
127	What is a buffer?	A buffer resists change in pH when small amounts of acid or base are added.
128	Name the major extracellular buffer system.	Bicarbonate buffer system.
129	What is acidosis?	A decrease in blood pH below normal.
130	What is alkalosis?	An increase in blood pH above normal.